SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I **_Bandaru Jayasri-192424301**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = **150 MB** = 150 × 1024 × 1024 = **157,286,400 bytes**
- Segment size = **1500 bytes**
- Data Link overhead = **40 bytes/frame**
- Bandwidth = **50 Mbps**

Step 1: Number of Segments

157,286,4001500=104,857.6\frac{157,286,400}{1500}=104,857.61500157,286,400
=104,857.6

Segments = 104,858

Since each segment becomes one frame,

Frames = 104,858

Step 2: Total Data Link Overhead

104,858×40=4,194,320 bytes104,858 \times 40 = 4,194,320 \text{
bytes}104,858×40=4,194,320 bytes

≈ **4 MB**

Step 3: Total Data Transmitted

157,286,400+4,194,320=161,480,720 bytes157,286,400+4,194,320=161,480,720\text{
bytes}157,286,400+4,194,320=161,480,720 bytes

Convert to bits:

161,480,720×8=1,291,845,760 bits161,480,720 \times 8=1,291,845,760\text{ bits}161,480,720×8=1,291,845,760 bits

Transmission time:
1,291,845,76050×106=25.84 s\frac{1,291,845,760}{50\times10^6}=25.84\text{ s}50×1061,291,845,760=25.84 s

Answer

- Segments = **104,858**
- Frames = **104,858**
- Data Link overhead = **4,194,320 bytes (≈4 MB)**
- Transmission time = **25.84 s**

Explanation

- **Transport Layer:** Segments data, performs flow control, error recovery, acknowledgements, and retransmissions.
- **Data Link Layer:** Creates frames, detects transmission errors using CRC, and ensures reliable node-to-node delivery.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given

- Window size = 6
- Total frames = 48
- Frame size = 1024 bytes

Number of Windows

486=8\frac{48}{6}=8648=8

Windows = 8

Retransmissions

1 frame lost in every window

8×1=88\times1=88×1=8 **Answer**

- Windows = **8**
- Retransmissions = **8**

Explanation

Flow control prevents sender overflow, matches receiver speed, reduces congestion, and improves throughput.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given

Fragments

120001480=8.11\frac{12000}{1480}=8.11148012000=8.11

Fragments = **9**

Header Overhead

9×20=180 bytes9\times20=180\text{ bytes}9×20=180 bytes

Answer

- Fragments = **9**
- Header overhead = **180 bytes**

Explanation

The Network Layer fragments packets, assigns fragment offsets and identification numbers, and reassembles them at the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Answer

- Windows = **8**
- Retransmissions = **8**

Explanation

Flow control prevents packet loss by regulating the transmission rate according to receiver capacity.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given

- File = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Checkpoints Before Failure

390 ÷ 60 = 6 checkpoints

(Checkpoints at 60,120,180,240,300,360 MB)

Retransmission

Last checkpoint = 360 MB

Failure = 390 MB

Retransmit:

390−360=30 MB390-360=30\text{ MB}390−360=30 MB

Answer

- Checkpoints = **6**
- Retransmitted data = **30 MB**

Explanation

Synchronization checkpoints allow transmission to resume from the latest checkpoint instead of restarting from the beginning.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given

- Original = 900 MB
- Compression = 40%
- Encryption overhead = 5%

Compressed Size

900×0.60=540 MB900\times0.60=540\text{ MB}900×0.60=540 MB

Encrypted Size

540×1.05=567 MB540\times1.05=567\text{ MB}540×1.05=567 MB

Percentage Reduction

900−567900×100=37%\frac{900-567}{900}\times100=37\%900900−567×100=37%

Answer

- Compressed size = **540 MB**
- Encrypted size = **567 MB**
- Total reduction = **37%**

Explanation

The Presentation Layer compresses data to reduce transmission time and encrypts it to ensure confidentiality.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given

- Data = 3 GB = 3072 MB
- Throughput = 120 Mbps
- Request size = 3 MB

Requests

30723=1024\frac{3072}{3}=102433072=1024

Transmission Time

3072 MB = 24,576 Mb

24,576120=204.8 s\frac{24,576}{120}=204.8\text{ s}12024,576=204.8 s

Answer

- Requests = **1024**
- Transmission time = **204.8 seconds**

Explanation

The Application Layer provides user services such as FTP commands, authentication, error reporting, and reliable file transfer.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Processing delay:

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data Link = 2 ms Physical

= 1 ms

Total Processing:

4+3+5+2+1=15 ms4+3+5+2+1=15\text{ ms}4+3+5+2+1=15 ms

Propagation = 18 ms Transmission

= 12 ms

Total Delay:

15+18+12=45 ms15+18+12=45\text{ ms}15+18+12=45 ms

Answer

Total end-to-end delay = **45 ms**

Explanation

- Application: User services
- Transport: Segmentation and reliability
- Network: Routing
- Data Link: Framing and error detection
- Physical: Bit transmission

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given

- Useful data = 80 MB

- Frame size = 1600 bytes
- Overhead = 80 bytes/frame

Useful payload/frame:
1600−80=1520 bytes1600-80=1520\text{ bytes}1600−80=1520 bytes

Useful data:

80 MB = 83,886,080 bytes

Frames

83,886,0801520=55,188.2\frac{83,886,080}{1520}=55,188.2152083,886,080=55,188.2

Frames = **55,189**

Total Overhead

55,189×80=4,415,120 bytes55,189\times80=4,415,120\text{ bytes}55,189×80=4,415,120 bytes

Efficiency

15201600×100=95%\frac{1520}{1600}\times100=95\%16001520×100=95%

Answer

- Frames = **55,189**
- Overhead = **4,415,120 bytes**
- Transmission efficiency = **95%**

Explanation

Protocol overhead consumes bandwidth, reducing effective throughput and increasing transmission time.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given

- Original = 240 MB
- Compression = 30%
- Segment = 1400 bytes
- Frame overhead = 60 bytes

Compressed Size

240×0.70=168 MB240\times0.70=168\text{ MB}240×0.70=168 MB

Compressed bytes:
168×1024×1024=176,160,768 bytes168\times1024\times1024=176,160,768\text{ bytes}168×1024×1024=176,160,768 bytes

Segments

176,160,7681400=125,829.12\frac{176,160,768}{1400}=125,829.121400176,160,768 =125,829.12

Segments = **125,830**

Protocol Overhead

125,830×60=7,549,800 bytes125,830\times60=7,549,800\text{ bytes}125,830×60=7,549,800 bytes

Final Data

176,160,768+7,549,800=183,710,568 bytes176,160,768+7,549,800=183,710,568\text{ bytes}176,160,768+7,549,800=183,710,568 bytes

≈ **175.2 MB**

Answer

- Compressed size = **168 MB**
- Segments = **125,830**
- Protocol overhead = **7,549,800 bytes**
- Final transmitted data = **183,710,568 bytes (≈175.2 MB)**

Explanation

- **Presentation Layer:** Compresses and encrypts data.
- **Transport Layer:** Segments data and provides reliable delivery.
- **Data Link Layer:** Frames data and detects errors.
- **Physical Layer:** Transmits bits over the communication medium.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand